

FREEMARKET · SENIOR DATA ENGINEER TAKE-HOME

A trustworthy, GBP-normalised money-flow network

Bronze → Silver → Gold in a single DuckDB file — from four raw sources to a graph-ready Gold dataset, with every number traceable back to an untouched source row.

12,982

transactions · Jul-Dec 2025

£26.15bn

GBP-normalised volume

1,376

network nodes

4,110

directed edges

101

tests gating the build

THE PROBLEM

Commercial and Finance can't answer a basic question

“Who is moving how much money with whom, and when — and what revenue did we earn on it?”

WHY IT'S HARD

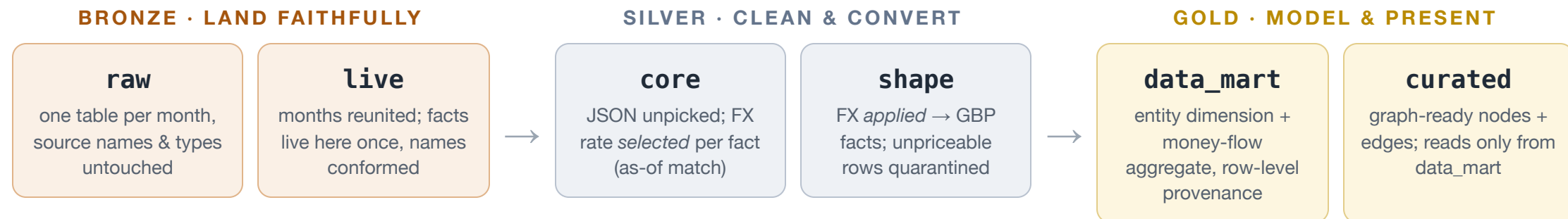
- Four raw extracts from different systems, **not built for analytics**: a 4-sheet Excel workbook and three differently-nested JSON files
- Amounts in **many currencies**; FX rates arrive as ~161k unsorted, time-versioned tuples
- The corporate hierarchy (group → legal entity → counterparty) is **implicit across sources**

WHAT WAS ASKED

- A **Medallion pipeline in DuckDB** — one local file, SQL + Python
- A Gold dataset driving a **directed money-flow network**: GBP volume, count and fee revenue per relationship
- **Sliceable** by month/year, **drillable** up and down the group ↔ company hierarchy

ARCHITECTURE

Three layers, six schemas, one file — per the team's build protocol



THE TOPOLOGY DECISION — CATALOG VS SCHEMA

DuckDB shares the `catalog.schema.table` namespace with Databricks, but a DuckDB catalog is a **physical file** — the “catalog per layer” idiom would need three attached files and break the single-file constraint. So layers map to **schemas**; if the rule were lifted, they lift mechanically to catalogs (`ATTACH 'bronze.duckdb'`) with schema names unchanged.

THE UNDERPINNING PRINCIPLE

Keep the raw fact pure. Facts land once in Bronze `live` ; FX attaches as a separate keyed table, never as columns bolted onto the fact. Lineage back to an untouched source row is what makes row and measure counts checkable at every layer boundary.

FX: every amount priced at its own instant — or not at all

- **Point-in-time as-of join:** each amount converts at the rate whose validity window spans its settlement instant (`Tx Date + Tx Time` ; fees at their date) — cross-currency totals only compare if each amount is priced at its own moment
- **One pure, isolated unit** (`src/fx.py`): sorts the unsorted rate points once, range-matches `validFrom ≤ t < validTill` by bisect; independently tested with neither DuckDB nor the 18 MB file
- **GBP is an explicit identity** (rate 1.0 — it has no series in the file), not a fudge
- **Full lineage:** every priced fact carries the `fx_rate_id` of the exact rate point that priced it

NEVER SILENTLY MIS-PRICED

Anything unpriceable is **quarantined with a stated reason** — kept and auditable, never dropped or null-filled:

<code>fx_out_of_coverage</code>	instant outside the file's window
<code>fx_unknown_currency</code>	no series for the currency
<code>fx_no_rate_point</code>	gap in the series at that instant
<code>fx_missing_currency</code>	row has no currency
<code>fx_missing_instant</code>	row has no settlement instant

Splitting “which rate” (`core`) from “apply the rate” (`shape`) separates the two ways FX can go wrong into two inspectable stages.

DATA QUALITY

Nothing vanishes silently — every exclusion classified, every boundary gated

WHAT THE DATA THREW UP	COUNT	VERDICT	WHY
Fees with no currency	42	quarantined	unpriceable (<code>fx_missing_currency</code>) — the entire quarantine ledger
Transaction IDs in both Deposit & Withdrawals	306	kept	distinct streams reusing IDs; fees key on (<code>link_id</code> , <code>fee_type</code>) so nothing fans out
Counterparties with no group/company key	1,363	kept	standalone by design → diamond nodes, exactly as the brief expects
Counterparties with a key	222	kept	all resolve deterministically to a group — zero orphans, no name-matching needed
Duplicate fact key within a stream	0	would fail loud	unique key asserted before pricing — a dup would inflate volume and fan out fees
Null amounts · out-of-period rows	0	quarantine · fail loud	null amount → quarantined like a bad currency; out-of-period → Bronze raises. Enforced, not just observed — the next drop can't slip a bad row through

Conservation gates the build: per-month rows sum to `live` · promoted + quarantined = input · the 12,982 transactions land in `curated.edge` uncounted-out · GBP totals unchanged Silver → Gold. Any breach turns `make test` red.

THE DELIVERABLE

Gold curated drives the graph directly — zero reshaping



make render → this star map, read straight from curated.node + curated.edge : focal group **Tech Key**, top-12 counterparts by volume, full period. ● circles = groups (group↔group flow is first-class — 449 such edges) ◆ diamonds = standalone counterparties · directed edges labelled £ volume · txn count · fee revenue (green in, orange out).

SLICE & DRILL

Any group, any period, any level of the hierarchy — one **WHERE** clause away

- Every edge carries `month` and `year` — the month grain is additive, so year and full-period views are plain **roll-ups by summation**
- Every edge carries both `focal_group_id` and `focal_company_id` (+ its name) — the fact is stored at the **finest company grain**, so the group view is the sum of its companies, and the drill is **recoverable, not lost**
- Proven in SQL (`notebooks/drill.ipynb`): star view → named legal entities → one company's own counterparts → **roll-up reconciles to the penny**

THE DRILL, CONCRETELY (KOBAMAYA)

Group grain: who does Koba2Maya, as a whole, move money with?

↓ **drill:** which of its 18 legal entities transacted — *two separate entities both named* “Operations Ltd” (£66m and £18m), “Group Holdings £6.7m”...

↓ **drill:** who did the £66m entity itself deal with?

↑ **roll up:** the group total equals the sum of its companies — *drill is lossless*.

Hierarchy comes from deterministic keys only (`dc_id` → `company` → `parentGroup`) — no name-matching, no guessing. Names duplicate across jurisdictions; the **id is the key**, the name is just the label.

IF THIS WERE A LIVE STREAM

The design already splits at the right seams — four concrete deltas

- 1 **Swap the ingestion source.** `build_raw_monthly` reads a sheet today; replace that one function with a stream consumer appending events to the Bronze `raw` month partition. Nothing downstream of `live` needs to know.
- 2 **Make Bronze writes append-only and idempotent.** `INSERT ... ON CONFLICT DO NOTHING` on the natural key, so at-least-once delivery and replays can't double-count.
- 3 **Process incrementally.** Watermark the new rows; `money_flow` is additive on its grain, so a micro-batch is an upsert into the affected month's aggregate, not a full rebuild.
- 4 **Refresh FX as a live dimension.** A late-arriving rate point re-prices rows quarantined as `fx_no_rate_point` — nothing stays stuck.

Unchanged: the FX unit, the JSON unpicking, the network model, the quarantine rules, the conservation checks — the transformation logic never cared whether a row came from a sheet or a stream.

KEY DECISIONS

The load-bearing choices — and the alternatives they beat

- **Attach FX, don't bolt it on** — the match lives in its own table keyed to the fact. Rejected: `fx_*` columns on the fact (mutates the raw fact, collapses two failure modes into one).
- **Quarantine is terminal** — Gold reads only promoted facts; no wrong or null number can reach the product. Rejected: null-filled GBP (silently poisons totals).
- **Money-flow at the finest grain** — company-level rows, group view as roll-up. Rejected: group grain (drill would be unrecoverable — `curated` reads only `data_mart`).
- **Enforce integrity, don't assume it** — duplicate keys fail the build, null amounts quarantine; no-ops on this drop, protection for the next one.
- **Deterministic counterpart** → **group resolution** (`group_id` , else `dc_id` → `company` → `group`) — zero orphans found, so no fragile name-matching anywhere.
- **One documented deviation** from the build protocol: singular table names (`live.deposit` , not `deposits`) — one naming convention across all six schemas; the protocol's substance (one consolidated table per stream) is unchanged.

Reproducible by construction, honest about the one unknown

TRUST

- `make pipeline` rebuilds the committed warehouse from scratch, deterministically — same totals every run
- **101 tests** gate the build: the pure FX unit, per-layer invariants, an end-to-end conservation spine, and a write-up completeness tripwire
- Row-level provenance (`source lists`) and FX lineage (`fx_rate_id`) on every Gold row

THE ONE STAKEHOLDER QUESTION

The reference snapshot is from a **different data drop** — its figures don't exist in this data, so it guided *shape*, not numbers. Whether any snapshot figure is a validated reference for a named group and period is the single question worth an email; everything else was resolvable from the data.

NEXT

- Incremental micro-batch ingestion + orchestrator (Dagster/Airflow)
- A Gold DQ-reporting view — the quarantine ledger as a product
- Click-to-expand drill UI on top of `curated`